复旦数学资料汇编
代数数论

来源：香蕉空间 Solution 资料库

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第一章　代数数论

1.1　试卷: 代数数论

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1. Judgement (Just write true or false)

   (1) For any Dirichlet character χ, the L-function $L_\chi(s)$ is holomorphic in the half plane $\mathrm{Re}\, s > 0$;

   (2) $\frac{\sqrt{3}+\sqrt{5}}{2}$ is an algebraic integer;

   (3) For any number field L, $x \in \mathcal{O}_L \Leftrightarrow N_{L/\mathbb{Q}}(x) \in \mathbb{Z}$;

   (4) The rational prime p such that $x^2 + 1 \equiv 0 \pmod{p}$ has solution has Dirichlet density $\frac{1}{2}$ in all the rational primes;

   (5) For any valuation field (K, v) and a finite algebraic extension L/K, v has a unique extension in L.

2. Denote $f(x) = x^3 - x^2 - 2x - 8$, $K = \mathbb{Q}[x]/f(x)$, α the image of x in K.

   (1) Prove: f is irreducible over $\mathbb{Q}$;

   (2) How many primes of $\mathcal{O}_K$ are over prime 2?

   (3) Compute $\mathrm{Disc}(1, \alpha, \alpha^2)$;

   (4) Prove that $\beta \triangleq \frac{\alpha^2-\alpha}{2}$ is integral over $\mathbb{Z}$;

   (5) Prove: $\{1, \alpha, \beta\}$ form an integral basis of $K/\mathbb{Q}$.

3. Let ζ_7 be the 7-th root of unity, K be the unique intermediate of $\mathbb{Q}(\zeta_7)/\mathbb{Q}$ such that $[K : \mathbb{Q}] = 3$.

   (1) Does there exist an embedding $\sigma : K \hookrightarrow \mathbb{C}$ such that the image of σ is not in $\mathbb{R}$?

   (2) Classify by module 7 which rational primes are ramified, inert and split in $\mathcal{O}_K$.

4. (1) Prove: The series

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{5} + \frac{1}{7} - \frac{1}{8} + \cdots$$

   converges conditionally in $\mathbb{R}$, and find its value;

   (2) In the p-adic number field $\mathbb{Q}_p$, prove that the series

$$\sum_{n=1}^{\infty} n \cdot n!, \quad \sum_{n=1}^{\infty} n^2 \cdot (n+1)!$$

   converge and find their values.

5. (1) Prove: Let p be an odd prime, then $\mathbb{Q}_p$ has no p-th root of unity other than 1.
   (2) Prove: $\mathbb{Q}_2$ has no 4-th root of unity other than ± 1.
   (3) Find all $m \in \mathbb{N}$ such that $\mathbb{Q}_p$ has all m-th roots of unity.
   (4) Prove: $\mathbb{Q}_p$ is not isomorphic to $\mathbb{Q}_q$ for $p \neq q$.
6. Let $\mathbb{F}_p((X))$ be the field of Laurent series over $\mathbb{F}_p$.
   (1) Does $\mathbb{F}_p((X))$ have an archimedean valuation?
   (2) Classify all valuations on $\mathbb{F}_p((X))$ under equivalence.